Solution of System of Linear Equations

: 10 hrs

1. Direct Method:
   a) Gauss- Elimination Method (with partial pivoting)
   b) Gauss-Jordan
   c) Factorization: Dolittle, Crout's, Choleskey's

⇒ Let the given system be

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$
$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

The augmented coefficient matrix is:

$$A/B = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right]$$

i> By Gauss - Elimination Method,
⇒ The augmented matrix is reduced to the form U/B' where U is the upper triangular matrix.

$$U/B' = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a'_{22} & a'_{23} & b'_2 \\ 0 & 0 & a''_{33} & b''_3 \end{array}\right]$$

Apply block substitution process.

ii> Gauss-Jordan Method:

⇒ The augmented matrix is reduced into the form I/B', where I is identity matrix.

$$\therefore I/B' = \left[\begin{array}{ccc|c} 1 & 0 & 0 & b_1' \\ 0 & 1 & 0 & b_2' \\ 0 & 0 & 1 & b_3' \end{array}\right]$$

$$\therefore x_1 = b_1' \; ; \; x_2 = b_2' \; ; \; x_3 = b_3'$$

Numericals:

1. Apply gauss elimination method with partial pivoting (if necessary) to solve:

$$10x - 7y + 3z + 5u = 6$$
$$-6x + 8y - z - 4u = 5$$
$$3x + y + 4z + 11u = 2$$
$$5x - 9y - 2z + 4u = 7$$

Sol The augmented coefficient matrix is:

$$A/B = \left[\begin{array}{cccc|c} 10 & -7 & 3 & 5 & 6 \\ -6 & 8 & -1 & -4 & 5 \\ 3 & 1 & 4 & 11 & 2 \\ 5 & -9 & -2 & 4 & 7 \end{array}\right]$$

Apply: $R_2 \to 5R_2 + 3R_1$; $R_3 \to 10R_3 - 3R_1$; $R_4 \to 2R_4 - R_1$, we get:

$$\sim \left[\begin{array}{cccc|c} 10 & -7 & 3 & 5 & 6 \\ 0 & 19 & 4 & -5 & 43 \\ 0 & 31 & 31 & 95 & 2 \\ 0 & -11 & -7 & 3 & 8 \end{array}\right]$$

Apply : $R_2 \leftrightarrow R_3$, we get;

$$\sim \left[\begin{array}{cccc|c} 10 & -7 & 3 & 5 & 6 \\ 0 & 31 & 31 & 95 & 2 \\ 0 & 19 & 4 & -5 & 43 \\ 0 & -11 & -7 & 3 & 8 \end{array}\right]$$

Apply : $R_3 \to 31R_3 - 19R_2$; $R_4 \to 31R_4 + 11R_2$, we get;

$$\sim \left[\begin{array}{cccc|c} 10 & -7 & 3 & 5 & 6 \\ 0 & 31 & 31 & 95 & 2 \\ 0 & 0 & -465 & -1960 & 1295 \\ 0 & 0 & 124 & 1138 & 270 \end{array}\right]$$

Apply : $R_4 \to 465R_4 + 124R_3$, we get;-

$$\sim \left[\begin{array}{cccc|c} 10 & -7 & 3 & 5 & 6 \\ 0 & 31 & 31 & 95 & 2 \\ 0 & 0 & -465 & -1960 & 1295 \\ 0 & 0 & 0 & 286130 & 286130 \end{array}\right]$$

Apply back substitution method,

$\because 286130u = 286130 \quad \therefore u = 1$

$\because -465z - 1960u = 1295$

$\Rightarrow -465z - 1960 \times 1 = 1295$

$\Rightarrow -465z = 1295 + 1960 = 3255$

$\therefore z = -7$

$\because 31y + 31z + 95u = 2$

$\Rightarrow 31y + 31 \times -7 + 95 \times 1 = 2$

$\Rightarrow 31y = 2 - 95 + 217$

$\Rightarrow y = 124/31$

$\therefore y = 4$

$\because 10x - 7y + 3z + 5u = 6$

$\Rightarrow 10x - 7\times 4 + 3\times -7 + 5\times 1 = 6$

$\Rightarrow 10x = 6 + 28 + 21 - 5$

$\Rightarrow x = 50/10$

$\therefore x = 5$

Thus the required solution of given system:

$x = 5;\ y = 4;\ z = -7;\ u = 1.$ Ans

2. Apply Gauss-Jordan method to solve,

$$2x + 5y + 7z = 52$$
$$2x + 1y - 1z = 0$$
$$x + y + z = 9$$

Sol The augmented coefficient matrix is:

$$A/B = \left[\begin{array}{ccc|c} 2 & 5 & 7 & 52 \\ 2 & 1 & -1 & 0 \\ 1 & 1 & 1 & 9 \end{array}\right]$$

Apply: $R_2 \to R_2 - R_1;\ R_3 \to 2R_3 - R_1$, we get:

$$\sim \left[\begin{array}{ccc|c} 2 & 5 & 7 & 52 \\ 0 & -4 & -8 & -52 \\ 0 & -3 & -5 & -34 \end{array}\right]$$

Apply: $R_1 \to 4R_1 + 5R_2;\ R_3 \to 4R_3 - 3R_2$, we get;

$$\sim \left[\begin{array}{ccc|c} 8 & 0 & -12 & -52 \\ 0 & -4 & -8 & -52 \\ 0 & 0 & 4 & 20 \end{array}\right]$$

Apply: $R_1 \to R_1 + 3R_3$; $R_2 \to R_2 + 2R_3$, we get;-

$$\sim \left[\begin{array}{ccc|c} 8 & 0 & 0 & 8 \\ 0 & -4 & 0 & -12 \\ 0 & 0 & 4 & 20 \end{array}\right]$$

Apply: $R_1 \to \frac{R_1}{8}$; $R_2 \to \frac{R_2}{-4}$; $R_3 \to \frac{R_3}{4}$, we get;-

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 5 \end{array}\right]$$

Thus, the required solution of the given system:

$\therefore\ x = 1$; $y = 3$; $z = 5$ Ans

iii) Factorization (Decomposition) Method:

⇒ Let the given system be

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

The matrix form of the given system:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\therefore AX = B \quad \text{———} \quad (1)$$

Let $A = L.U$

$$= \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} . \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

a> If $l_{11} = l_{22} = l_{33} = 1$, then the method is called Dolittle factorization.

b> If $u_{11} = u_{22} = u_{33} = 1$, then the method is called Court's factorization.

E.g. If we take $l_{11} = l_{22} = l_{33} = 1$, then

$$A = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} . \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Then multiply RHS and compare the corresponding elements with the matrix A.

From (1),

$$L \cdot U \cdot X = B \quad \text{——— (2)}$$

put $U \cdot X = Z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$ ——— (3)

From (2),

$$L \cdot Z = B$$

We obtain the value of z. Therefore, the solution is obtained from eqn. (3).

Eg: If we take $u_{11} = u_{22} = u_{33} = 1$, then

$$A = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \cdot \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

Then multiply ~~the~~ RHS and compare the corresponding elements with the matrix A.

From (1),

$$L \cdot U \cdot X = B \quad \text{——— (2)}$$

put $UX = Z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$ ——— (3)

From (2),

$$LZ = B$$

We obtain the value of z. Therefore, the solution is obtained from eqn. (3).

Numericals:

1. Using Dolittle factorization method, solve:

$$3x+2y+7z=4$$
$$2x+3y+z=5$$
$$3x+4y+z=7$$

Sol The matrix form of the given system is:

$$\begin{bmatrix}3 & 2 & 7\\ 2 & 3 & 1\\ 3 & 4 & 1\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}4\\ 5\\ 7\end{bmatrix}$$

$$\therefore \quad A.X=B \quad \text{——} \quad (1)$$

Using Dolittle factorization method,

$$A=L.U$$

$$\Rightarrow A=\begin{bmatrix}1 & 0 & 0\\ l_{21} & 1 & 0\\ l_{31} & l_{32} & 1\end{bmatrix}.\begin{bmatrix}u_{11} & u_{12} & u_{13}\\ 0 & u_{22} & u_{23}\\ 0 & 0 & u_{33}\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}3 & 2 & 7\\ 2 & 3 & 1\\ 3 & 4 & 1\end{bmatrix}=\begin{bmatrix}u_{11} & u_{12} & u_{13}\\ l_{21}u_{11} & l_{21}u_{12}+u_{22} & l_{21}u_{13}+u_{23}\\ l_{31}u_{11} & l_{31}u_{12}+l_{32}u_{22} & l_{31}u_{13}+l_{32}u_{23}+u_{33}\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}3 & 2 & 7\\ 2 & 3 & 1\\ 3 & 4 & 1\end{bmatrix}=\begin{bmatrix}u_{11} & u_{12} & u_{13}\\ l_{21}u_{11} & l_{21}u_{12}+u_{22} & l_{21}u_{13}+u_{23}\\ l_{31}u_{11} & l_{31}u_{12}+l_{32}u_{22} & l_{31}u_{13}+l_{32}u_{23}+u_{33}\end{bmatrix}$$

Equating corresponding matrices, we get;

From 1^{st} row,

$$u_{11}=3;\quad u_{12}=2;\quad u_{13}=7$$

From 2^{nd} row,

$$l_{21}u_{11}=2 \Rightarrow l_{21}=2/3$$

$l_{21} u_{12} + u_{22} = 3$

$\Rightarrow 2/3 \times 2 + u_{22} = 3$

$\Rightarrow u_{22} = 3 - 4/3 = \frac{9-4}{3} = \frac{5}{3}$

$l_{21} u_{13} + u_{23} = 1$

$\Rightarrow \frac{2}{3} \times 7 + u_{23} = 1$

$\Rightarrow u_{23} = 1 - \frac{14}{3} = -\frac{11}{3}$

From 3^{rd} row,

$l_{31} u_{11} = 3$

$\Rightarrow l_{31} \times 3 = 3$

$\therefore l_{31} = 1$

$l_{31} u_{12} + u_{32} u_{22} = 4$

$\Rightarrow 1 \times 2 + u_{32} \times \frac{5}{3} = 4$

$\Rightarrow u_{32} \times \frac{5}{3} = 4 - 2 = 2$

$\Rightarrow u_{32} = \frac{6}{5}$

$l_{31} u_{13} + l_{32} u_{23} + u_{33} = 1$

$\Rightarrow 1 \times 7 + \frac{6}{5} \times \frac{-11}{3} + u_{33} = 1$

$\Rightarrow u_{33} + 7 - \frac{22}{5} = 1$

$\Rightarrow u_{33} = 1 + \frac{22}{5} - 7 = \frac{22}{5} - 6$

$\Rightarrow u_{33} = \frac{22-30}{5}$

$\therefore u_{33} = -8/5$

Hence,

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1 & 6/5 & 1 \end{bmatrix}; \quad U = \begin{bmatrix} 3 & 2 & 7 \\ 0 & 5/3 & -11/3 \\ 0 & 0 & -8/5 \end{bmatrix}$$

From (1),

$$L \cdot U \cdot X = B \quad \text{———} (2)$$

Put $U \cdot X = Z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$ ——— (3)

From (2),

$L \cdot Z = B$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1 & 6/5 & 1 \end{bmatrix} \cdot \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} z_1 \\ \frac{2}{3}z_1 + z_2 \\ z_1 + \frac{6}{5}z_2 + z_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 7 \end{bmatrix}$$

Equating corresponding matrices, we get :-

From 1st row; $z_1 = 4$

From 2nd row;

$$\frac{2z_1}{3} + z_2 = 5$$

$$\Rightarrow z_2 = 5 - \frac{2}{3} \times 4$$

$$\therefore z_2 = \frac{15-8}{3} = \frac{7}{3}$$

From 3rd row,

$$z_1 + \frac{6}{5} z_2 + z_3 = 7$$

$$\Rightarrow 4 + \frac{6}{5} \times \frac{7}{3} + z_3 = 7$$

$$\Rightarrow 4 + \frac{14}{5} + z_3 = 7$$

$$\Rightarrow z_3 = 3 - \frac{14}{5} = \frac{15-14}{5}$$

$$\therefore z_3 = \frac{1}{5}$$

From (3),

$$\begin{bmatrix} 3 & 2 & 7 \\ 0 & 5/3 & -11/3 \\ 0 & 0 & -8/5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 7/3 \\ 1/5 \end{bmatrix}$$

Apply back substitution method,

From 1st row; $-\frac{8}{5}z = 1/5$ $\therefore z = -1/8$

From 2nd row;

$$\frac{5}{3}y - \frac{11}{3}z = \frac{7}{3}$$
$$\Rightarrow 5y - 11z = 7$$
$$\Rightarrow 5y - 11 \times -\frac{1}{8} = 7$$
$$\Rightarrow 5y = 7 - \frac{11}{8} = \frac{56-11}{8}$$
$$\Rightarrow 5y = \frac{45}{8}$$
$$\therefore y = 9/8$$

from 3rd row,

$$3x + 2y + 7z = 4$$
$$\Rightarrow 3x + 2 \times \frac{9}{8} + 7 \times -\frac{1}{8} = 4$$
$$\Rightarrow 3x = 4 - \frac{18}{8} + \frac{7}{8} = \frac{32-18+7}{8}$$
$$\Rightarrow 3x = \frac{21}{8}$$
$$\therefore x = 7/8$$

Thus by Dolittle factorization method, the solution of the given system is;

$x = 7/8$; $y = 9/8$; $z = -1/8$ Ans

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2. Using Court's factorization method, solve :

$$3x + y + 4z = 12$$
$$8x - 3y + 2z = 20$$
$$4x + 11y - z = 33$$

Sol: The matrix form of the given ~~equ~~ system is:

$$\begin{bmatrix} 3 & 1 & 4 \\ 8 & -3 & 2 \\ 4 & 11 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 20 \\ 33 \end{bmatrix}$$

$$\therefore A \cdot X = B \text{ ——— (1)}$$

Using Crout's factorization method, we get;

$$A = L \cdot U$$

$$\Rightarrow A = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \cdot \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 1 & 4 \\ 8 & -3 & 2 \\ 4 & 11 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12} + l_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31} & l_{31}u_{12} + l_{32} & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{bmatrix}$$

Equating Corresponding matrices, we get;

from 1st ~~row~~ column; $l_{11} = 3$; $l_{21} = 8$; $l_{31} = 4$

From 2nd ~~row~~ column;

$$l_{11}u_{12} = 1 \Rightarrow u_{12} = \frac{1}{3}$$

$$l_{21}u_{12} + l_{22} = -3$$

$$\Rightarrow 8 \times \frac{1}{3} + l_{22} = -3$$

$$\Rightarrow l_{22} = -3 - \frac{8}{3} = \frac{-17}{3}$$

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$l_{31}\, u_{12} + l_{32} = 11$

$\Rightarrow 4 \times \frac{1}{3} + l_{32} = 11$

$\therefore l_{32} = 11 - 4/3 = 29/3$

From 3^{rd} column,

$l_{11} u_{13} = 4 \qquad \therefore u_{13} = 4/3$

$l_{21} u_{13} + l_{22} . u_{23} = 2$

$\Rightarrow 8 \times 4/3 - 17/3\, u_{23} = 2$

$\Rightarrow \frac{32}{3} - 2 = \frac{17}{3} u_{23}$

$\therefore u_{23} = 26/17$

$l_{31} u_{13} + l_{32} u_{23} + l_{33} = -1$

$\Rightarrow 4 \times \frac{4}{3} + \frac{29}{3} \times \frac{26}{17} + l_{33} = -1$

$\Rightarrow l_{33} = -1 - \frac{16}{3} - \frac{754}{51}$

$\therefore l_{33} = -\frac{359}{17}$

Hence,

$$L = \begin{bmatrix} 3 & 0 & 0 \\ 8 & -17/3 & 0 \\ 4 & 29/3 & -359/17 \end{bmatrix} ; \quad U = \begin{bmatrix} 1 & 1/3 & 4/3 \\ 0 & 1 & 26/17 \\ 0 & 0 & 1 \end{bmatrix}$$

From (1),

$L . U . X = B \quad \text{——} \ (2)$

put $U . X = Z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \quad \text{———} \ (3)$

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2. From (2),

$$L \cdot Z = B$$

$$\Rightarrow \begin{bmatrix} 3 & 0 & 0 \\ 8 & -17/3 & 0 \\ 4 & 29/3 & -359/17 \end{bmatrix} \cdot \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 20 \\ 33 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3z_1 \\ 8z_1 - \frac{17}{3}z_2 \\ 4z_1 + \frac{29}{3}z_2 - \frac{359}{17}z_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 20 \\ 33 \end{bmatrix}$$

Equating corresponding matrices, we get;

From 1st row,

$$3z_1 = 12 \quad \therefore \quad z_1 = 4$$

From 2nd row,

$$8z_1 - \frac{17}{3}z_2 = 20$$

$$\Rightarrow 8 \times 4 - 20 = \frac{17}{3}z_2$$

$$\Rightarrow \frac{12 \times 3}{17} = z_2$$

$$\therefore z_2 = 21/17$$

From 3rd row,

$$4z_1 + \frac{29}{3}z_2 - \frac{359}{17}z_3 = 33$$

$$\Rightarrow 4 \times 4 + \frac{29}{3} \times \frac{21}{17} - \frac{359}{17}z_3 = 33$$

$$\Rightarrow 16 - 33 + \frac{29}{3} \times \frac{21}{17} = \frac{359}{17}z_3$$

$$\Rightarrow -\frac{86}{17} = \frac{359}{17}z_3$$

$$\therefore z_3 = -86/359$$

from (8),

$$\begin{bmatrix} 1 & 1/3 & 4/3 \\ 0 & 1 & 26/17 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 21/17 \\ -86/359 \end{bmatrix}$$

Apply back substitution method, we get;

$$z = -\frac{86}{359}$$

$$y + \frac{26}{17} z = \frac{21}{17}$$

$$\Rightarrow y = \frac{21}{17} - \frac{26}{17} \times \frac{-86}{359}$$

$$\therefore y = \frac{575}{359}$$

$$x + \frac{1}{3} y + \frac{4}{3} z = 4$$

$$\Rightarrow x + \frac{1}{3} \times \frac{575}{359} + \frac{4}{3} \times \frac{-86}{359} = 4$$

$$\Rightarrow x + \frac{77}{359} = 4$$

$$\therefore x = \frac{1359}{359}$$

Hence the required solution of given system is:

$$x = \frac{1359}{359} \;;\; y = \frac{575}{359} \;;\; z = \frac{-86}{359}.$$

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ii> Choleskey's Factorization Method:

→ If the coefficient matrix A is symmetric. Then,

$$A = L \cdot L^T$$

where,

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

$$L^T = \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

The given in the matrix form is:

$$AX = B \quad \text{———} \quad (1)$$

$$\Rightarrow L \cdot L^T X = B \quad \text{———} \quad (2)$$

put $L^T X = Z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$ ——— (3)

From (2) & (3), we get;

$$\therefore L \cdot Z = B$$

Numericals:

1. Factorize the given matrix using choleskey's method.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix}$$

Soln Here,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix}$$

$$\therefore A' = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix}$$

$\because A' = A$. Thus A is a symmetric matrix.

By choleskey's method,

$A = L \cdot L^T$

$$\Rightarrow A = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \cdot \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix} = \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{31}l_{11} \\ l_{11}l_{21} & l_{21}^2+l_{22}^2 & l_{21}l_{31}+l_{22}l_{32} \\ l_{11}l_{31} & l_{21}l_{31}+l_{22}l_{32} & l_{31}^2+l_{32}^2+l_{33}^2 \end{bmatrix}$$

Equating corresponding matrices, we get;-

$l_{11}^2 = 1$ $\qquad \therefore l_{11} = 1$

$l_{11}l_{21} = 2$ $\qquad \therefore l_{21} = 2$

$l_{11}l_{31} = 3$ $\qquad \therefore l_{31} = 3$

$l_{21}^2 + l_{22}^2 = 8 \Rightarrow 2^2 + l_{22}^2 = 8 \Rightarrow l_{22}^2 = 8-4 = 4$ $\qquad \therefore l_{22} = 2$

$l_{21}l_{31} + l_{22}l_{32} = 22$

$\Rightarrow 2 \cdot 3 + 2 \cdot l_{32} = 22$

$\Rightarrow 2 l_{32} = 22-6 = 16$ $\qquad \therefore l_{32} = 8$

$l_{31}^2 + l_{32}^2 + l_{33}^2 = 82$

$\Rightarrow l_{33}^2 + 3^2 + 8^2 = 82$

$\Rightarrow l_{33}^2 = 82 - 73 = 9$ $\qquad \therefore l_{33} = 3$

Now,

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix} ; L^T = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 8 \\ 0 & 0 & 3 \end{bmatrix}$$ Ans

2. Using Choleskey's factorization method, solve:

$$4x+y+z=7$$
$$x+5y+2z=3$$
$$x+2y+3z=1$$

Sol^n Here,

$$4x+y+z=7$$
$$x+5y+2z=3$$
$$x+2y+3z=1$$

The matrix form of the given system is:

$$\begin{bmatrix} 4 & 1 & 1 \\ 1 & 5 & 2 \\ 1 & 2 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

$$\therefore \quad A \cdot X = B \quad \text{———} (1)$$

Now,

$$A' = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 5 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

$\because A' = A$. So A is a symmetric matrix.

By choleskey's factorization method,

$$A = L \cdot L^T$$

$$\Rightarrow \begin{bmatrix} 4 & 1 & 1 \\ 1 & 5 & 2 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \cdot \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & 1 & 1 \\ 1 & 5 & 2 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{21}l_{11} & l_{21}^2+l_{22}^2 & l_{21}l_{31}+l_{22}l_{32} \\ l_{31}l_{11} & l_{31}l_{21}+l_{32}l_{22} & l_{31}^2+l_{32}^2+l_{33}^2 \end{bmatrix}$$

Equating corresponding matrices, we get;

$l_{11}^2 = 4$ $\quad\therefore l_{11} = 2$

$l_{11}l_{21} = 1$ $\quad\therefore l_{21} = 1/2$

$l_{31}l_{11} = 1$ $\quad\therefore l_{31} = 1/2$

$l_{21}^2 + l_{22}^2 = 5 \Rightarrow \cancel{1}/4 + l_{22}^2 = 5 \Rightarrow l_{22}^2 = 5 - 1/4 \quad \therefore l_{22} = \frac{\sqrt{19}}{2}$

$l_{21} l_{31} + l_{22} l_{32} = 2$

$\Rightarrow \frac{1}{2} \times \frac{1}{2} + \frac{\sqrt{19}}{2} \times l_{32} = 2 \Rightarrow 1 + 2\sqrt{19}\, l_{32} = 8 \quad \therefore l_{32} = \frac{7}{2\sqrt{19}}$

$l_{31}^2 + l_{32}^2 + l_{33}^2 = 3$

$\Rightarrow (1/2)^2 + \left(\frac{7}{2\sqrt{19}}\right)^2 + l_{33}^2 = 3 \Rightarrow l_{33}^2 = 3 - 1/4 - \frac{49}{4 \times 19} =$

$\Rightarrow l_{33} = \frac{2\sqrt{190}}{19}$

Now,

$$L = \begin{bmatrix} 2 & 0 & 0 \\ 1/2 & \sqrt{19}/2 & 0 \\ 1/2 & \frac{7}{2\sqrt{19}} & \frac{2\sqrt{190}}{19} \end{bmatrix} ; \quad L^T = \begin{bmatrix} 2 & 1/2 & 1/2 \\ 0 & \sqrt{19}/2 & \frac{7}{2\sqrt{19}} \\ 0 & 0 & \frac{2\sqrt{190}}{19} \end{bmatrix}$$

From (1),

$$L \cdot L^T X = B \quad \text{———— (2)}$$

put $L^T \cdot X = Z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$ ———— (3)

From (2) and (3),

$$L \cdot Z = B$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & 0 \\ 1/2 & \sqrt{19}/2 & 0 \\ 1/2 & 7/2\sqrt{19} & \frac{2\sqrt{190}}{19} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2z_1 \\ \frac{1}{2} z_1 + \frac{\sqrt{19}}{2} z_2 \\ \frac{1}{2} z_1 + \frac{7}{2\sqrt{19}} z_2 + \frac{2\sqrt{190}}{19} z_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

Apply back substitution method,

$2 z_1 = 7$

$\therefore z_1 = 7/2$

$\frac{1}{2}z_1 + \frac{\sqrt{19}}{2} z_2 = 3 \Rightarrow \frac{1}{2} \times \frac{7}{2} + \frac{\sqrt{19}}{2} z_2 = 3$

$\Rightarrow \frac{\sqrt{19}}{2} z_2 = 3 - \frac{7}{4} = \frac{5}{4} \quad \therefore \quad z_2 = \frac{5}{2\sqrt{19}}$

$\frac{1}{2}z_1 + \frac{7}{2\sqrt{19}} z_2 + \frac{2\sqrt{190}}{19} z_3 = 1$

$\Rightarrow \frac{1}{2} \times \frac{7}{2} + \frac{7}{2\sqrt{19}} \times \frac{5}{2\sqrt{19}} + \frac{2\sqrt{190}}{19} z_3 = 1$

$\Rightarrow \frac{2\sqrt{190}}{19} z_3 = 1 - \frac{7}{4} - \frac{35}{76} = -\frac{23}{19}$

$\Rightarrow z_3 = \frac{-23}{2\sqrt{190}}$

from (3),

$L^T x = z$

$$\Rightarrow \begin{bmatrix} 2 & \frac{1}{2} & \frac{1}{2} \\ 0 & \sqrt{19}/2 & 7/2\sqrt{19} \\ 0 & 0 & 2\sqrt{190}/19 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7/2 \\ 5/2\sqrt{19} \\ -23/2\sqrt{190} \end{bmatrix}$$

Apply back substitution method,

$\frac{2\sqrt{190}}{19} z = -23/2\sqrt{190}$

$\therefore z = \frac{-23}{2\sqrt{190}} \times \frac{19}{2\sqrt{190}} = -0.575$

$\sqrt{19}/2\, y + 7/2\sqrt{19}\, z = 5/2\sqrt{19}$

$\Rightarrow \frac{\sqrt{19}}{2} \times y + \frac{7}{2\sqrt{19}} \times -0.575 = \frac{5}{2\sqrt{19}}$

$\Rightarrow \frac{\sqrt{19}}{2} y = \frac{5}{2\sqrt{19}} + 0.462$

$\Rightarrow y = \frac{1.0355 \times 2}{\sqrt{19}} \quad \therefore y = 0.475$

$2x + \frac{1}{2}y + \frac{1}{2}z = 7/2 \Rightarrow 4x + y + z = 7 \Rightarrow 4x + 0.475 - 0.575 = 7$

$\Rightarrow 4x = 7 + 0.1 \Rightarrow x = 7.1/4 \quad \therefore x = 1.775$

Thus the solution of given system of linear equation is:

$x = 1.775$; $y = 0.475$; $z = -0.575$ Ans

* Inverse of a matrix using Gauss-Jordan Method

Let,

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$A/I = \left[\begin{array}{ccc|ccc} a_{11} & a_{12} & a_{13} & 1 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 & 1 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 & 1 \end{array}\right]$$

change this matrix in the form

$$I/A' = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & a'_{11} & a'_{12} & a'_{13} \\ 0 & 1 & 0 & a'_{21} & a'_{22} & a'_{23} \\ 0 & 0 & 1 & a'_{31} & a'_{32} & a'_{33} \end{array}\right]$$

Then,

$$A^{-1} = A' = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ a'_{21} & a'_{22} & a'_{23} \\ a'_{31} & a'_{32} & a'_{33} \end{bmatrix}$$

Numericals:

Q.1. Find the inverse of the matrix using Gauss-Jordan method.

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 2 & 3 \\ 3 & 4 & 2 \end{bmatrix}$$

Soln Here,

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 2 & 3 \\ 3 & 4 & 2 \end{bmatrix}$$

Now,

$$A/I = \left[\begin{array}{ccc|ccc} 2 & 3 & 4 & 1 & 0 & 0 \\ 4 & 2 & 3 & 0 & 1 & 0 \\ 3 & 4 & 2 & 0 & 0 & 1 \end{array}\right]$$

Apply: $R_2 \to R_2 - 2R_1$; $R_3 \to 2R_3 - 3R_1$, we get;-

$$\sim \left[\begin{array}{ccc|ccc} 2 & 3 & 4 & 1 & 0 & 0 \\ 0 & -4 & -5 & -2 & 1 & 0 \\ 0 & -1 & -8 & -3 & 0 & 2 \end{array}\right]$$

Apply: $R_1 \to 4R_1 + 3R_2$; $R_3 \to 4R_3 - R_2$, we get;-

$$\sim \left[\begin{array}{ccc|ccc} 8 & 0 & 1 & -2 & 3 & 0 \\ 0 & -4 & -5 & -2 & 1 & 0 \\ 0 & 0 & -27 & -10 & -1 & 8 \end{array}\right]$$

Apply: $R_1 \to 27R_1 + R_3$; $R_2 \to 27R_2 - 5R_3$, we get;-

$$\sim \left[\begin{array}{ccc|ccc} 216 & 0 & 0 & -64 & 80 & 8 \\ 0 & -108 & 0 & -4 & 32 & -40 \\ 0 & 0 & -27 & -10 & 1 & 8 \end{array}\right]$$

Apply: $R_1 \to R_1/216$; $R_2 \to R_2/-108$; $R_3 \to R_3/-27$, we get;

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -8/27 & 10/27 & 1/27 \\ 0 & 1 & 0 & 1/27 & -8/27 & 10/27 \\ 0 & 0 & 1 & 10/27 & 1/27 & -8/27 \end{array}\right]$$

Now,
$$A^{-1} = \begin{bmatrix} -8/27 & 10/27 & 1/27 \\ 1/27 & -8/27 & 10/27 \\ 10/27 & 1/27 & -8/27 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{27}\begin{bmatrix} -8 & 10 & 1 \\ 1 & -8 & 10 \\ 10 & 1 & -8 \end{bmatrix}$$ Ans

Q. Find A^{-1}:

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Sol: Here,

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Now,

$$A/I = \left[\begin{array}{ccc|ccc} 3 & 2 & 1 & 1 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array}\right]$$

Apply: $R_2 \to 3R_2 - 2R_1$; $R_3 \to 3R_3 - R_1$; we get:-

$$\sim \left[\begin{array}{ccc|ccc} 3 & 2 & 1 & 1 & 0 & 0 \\ 0 & 5 & 4 & -2 & 3 & 0 \\ 0 & 4 & 8 & -1 & 0 & 3 \end{array}\right]$$

Apply: $R_1 \to 5R_1 - 2R_2$; $R_3 \to 5R_3 - 4R_2$, we get;

$$\sim \left[\begin{array}{ccc|ccc} 15 & 0 & -3 & 9 & -6 & 0 \\ 0 & 5 & 4 & -2 & 3 & 0 \\ 0 & 0 & 24 & 3 & -12 & 15 \end{array}\right]$$

Apply: $R_1 \to 8R_1 + R_3$; $R_2 \to 6R_2 - R_3$, we get:

$$\sim \left[\begin{array}{ccc|ccc} 120 & 0 & 0 & 75 & -60 & 15 \\ 0 & 30 & 0 & -15 & 30 & -15 \\ 0 & 0 & 24 & 3 & -12 & 15 \end{array}\right]$$

Apply: $R_1 \to R_1/120$; $R_2 \to R_2/30$; $R_3 \to R_3/24$, we get

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5/8 & -1/2 & 1/8 \\ 0 & 1 & 0 & -1/2 & 1 & -1/2 \\ 0 & 0 & 1 & 1/8 & -1/2 & 5/8 \end{array}\right]$$

$$\therefore A^{-1} = \begin{bmatrix} 5/8 & -1/2 & 1/8 \\ -1/2 & 1 & -1/2 \\ 1/8 & -1/2 & 5/8 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{8}\begin{bmatrix} 5 & -4 & 1 \\ -4 & 8 & -4 \\ 1 & -4 & 5 \end{bmatrix}$$

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Ill Condition System:

→ A system of linear equation is said to be ill condition if the determinant of the coefficient ~~of~~ matrix is nearly zero.

→ Ill condition system is very sensitive to the round-off error.

→ That is ~~a~~ a small change in the coefficient or constant term produces a large change in the solution set.

→ The graph of such system is nearly parallel.

Example;

$$x - 2y = -2$$
$$0.45x - 0.91y = -1$$

Sol

∴ Determinant $= \begin{vmatrix} 1 & -2 \\ 0.45 & -0.91 \end{vmatrix} = -0.01 \approx 0$

on solving,

∴ $(x, y) = (18, 10)$

Another example;

$$2x_1 + x_2 = 25$$
$$2.001x_1 + x_2 = 25.01$$

Sol. Here,

$$\begin{vmatrix} 2 & 1 \\ 2.001 & 1 \end{vmatrix} = -0.001 \approx 0$$

The solution of the system is

$$(x_1, x_2) = (10, 5)$$

Again, if we write,

$$2x_1 + x_2 = 25$$
$$2.001x_1 + x_2 = 25$$

The solution of the system is

$$(x_1, x_2) = (0, 25).$$

2. Indirect (Iterative) Methods:

a> Jacobi - Method

b> Gauss Seidel Method

c> Relaxation Method

⇒ Let, $a_1x + b_1y + c_1z = d_1$

$a_2x + b_2y + c_2z = d_2$

$a_3x + b_3y + c_3z = d_3$

The system should be in diagonally dominant form.

i.e. $|a_1| > |b_1| + |c_1|$

$|b_2| > |a_2| + |c_2|$

$|c_3| > |a_3| + |b_3|$

Then solving for x, y and z respectively.

$$x = \frac{1}{a_1}(d_1 - b_1y - c_1z) \quad \text{——} (1)$$

$$y = \frac{1}{b_2}(d_2 - a_2x - c_2z) \quad \text{——} (2)$$

$$z = \frac{1}{c_3}(d_3 - a_3x - b_3y) \quad \text{——} (3)$$

Let $x_0 = y_0 = z_0 = 0$ be the initial guess.

a> Jacobi Method:

* Iteration 1:

From (1), $x^{(1)} = \frac{1}{a_1}(d_1 - b_1y_0 - c_1z_0)$

From (2), $y^{(1)} = \frac{1}{b_2}(d_2 - a_2x_0 - c_2z_0)$

From (3), $z^{(1)} = \frac{1}{c_3}(d_3 - a_3x_0 - b_3y_0)$

* **Iteration 2:**

$$x^{(2)} = \frac{1}{a_2}(d_1 - b_1 y^{(1)} - c_1 z^{(1)})$$

$$y^{(2)} = \frac{1}{b_2}(d_2 - a_2 x^{(1)} - c_2 z^{(1)})$$

$$z^{(2)} = \frac{1}{c_3}(d_3 - a_3 x^{(1)} - b_3 y^{(1)})$$ and so on.

b> **Gauss-Seidel Method:**

* **Iteration 1:**

$$x^{(1)} = \frac{1}{a_1}(d_1 - b_1 y_0 - c_1 z_0)$$

$$y^{(1)} = \frac{1}{b_2}(d_2 - a_2 x^{(1)} - c_2 z_0)$$

$$z^{(1)} = \frac{1}{c_3}(d_3 - a_3 x^{(1)} - b_3 y^{(1)})$$

* **Iteration 2:**

$$x^{(2)} = \frac{1}{a_1}(d_1 - b_1 y^{(1)} - c_1 z^{(1)})$$

$$y^{(2)} = \frac{1}{b_2}(d_2 - a_2 x^{(2)} - c_2 z^{(1)})$$

$$z^{(2)} = \frac{1}{c_3}(d_3 - a_3 x^{(2)} - b_3 y^{(2)})$$ and so on.

Numericals:

1. Solve:

$$x + 5y - z = 10$$
$$x + y + 8z = 20$$
$$4x + 2y + z = 14$$

i> using Jacobi method

ii> Gauss-Seidel Method.

Sol^n Here,

$$x + 5y - z = 10$$
$$x + y + 8z = 20$$
$$4x + 2y + z = 14$$

Rearrange the given equation in the diagonally dominant form, i.e.

$$4x+2y+z=14$$
$$x+5y-z=10$$
$$x+y+8z=20$$

Now,

$$x=\frac{1}{4}(14-2y-z) \text{ ——— } (1)$$

$$y=\frac{1}{5}(10-x+z) \text{ ——— } (2)$$

$$z=\frac{1}{8}(20-x-y) \text{ ——— } (3)$$

Let $x_0=y_0=z_0=0$ be the initial guess.

i> By Jacobi-Method

It no.	x	y	z
1.	3.5	2	2.5
2.	1.88	1.8	1.81
3.	2.15	1.99	2.04
4.	2.00	1.99	1.98
5.	2.01	2.00	2.00

ii> By Gauss-Seidel Method.

It no.	x	y	z
1.	3.5	1.3	1.9
2.	2.38	1.90	1.97
3.	2.06	1.98	2.00
4.	2.01	2.00	2.00

2. Using Gauss-Seidel method, solve:

$$27x+6y-z=85$$
$$x+y+54z=110$$
$$6x+15y+2z=72$$

Sol Here,

$$27x+6y-z=85$$
$$x+y+54z=110$$
$$6x+15y+2z=72$$

Rearrange the given equation in the diagonally dominant form. i.e.

$$27x + 6y - z = 85$$
$$6x + 15y + 2z = 72$$
$$x + y + 54z = 110$$

Now,

$$x = \frac{1}{27}(85 - 6y + z) \quad \text{---} (1)$$

$$y = \frac{1}{15}(72 - 6x - 2z) \quad \text{---} (2)$$

$$z = \frac{1}{54}(110 - x - y) \quad \text{---} (3)$$

~~Iteration~~

Let $x_0 = y_0 = z_0 = 0$ be the initial guess.

By Gauss-Seidel Method,

Iteration Table:

It no	x	y	z
1.	3.15	3.54	1.91
2.	2.43	3.57	1.93
3.	2.43	3.57	1.93

From the above solution,

$x = 2.43$; $y = 3.57$; $z = 1.93$ Ans

c) Successive Over Relaxation (SOR) Method:

⇒ Let,
$$a_1x+b_1y+c_1z=d_1$$
$$a_2x+b_2y+c_2z=d_2$$
$$a_3x+b_3y+c_3z=d_3$$
be a diagonally dominant system.

The residuals are obtained by
$$R_x = d_1 - a_1x - b_1y - c_1z$$
$$R_y = d_2 - a_2x - b_2y - c_2z$$
$$R_z = d_3 - a_3x - b_3y - c_3z$$

Let $x=y=z=0$ be the initial guess.

Operation Table:

	R_x	R_y	R_z
$\delta x=1$	$-a_1$	$-a_2$	$-a_3$
$\delta y=1$	$-b_1$	$-b_2$	$-b_3$
$\delta z=1$	$-c_1$	$-c_2$	$-c_3$

Relaxation Table:

	R_x	R_y	R_z
$x=y=z=0$	d_1	$\boxed{d_2}$	d_3
$\delta y=m_1$	$\boxed{A_1}$	A_2	A_3
$\delta z=m_2$	B_1	B_2	B_3

Let d_2 be the largest
$$d_2/b_2 = m_1$$
$$R_x = d_1 - (b_1)m_1 = A_1$$
$$R_y = d_2 - (b_2)m_1 = A_2$$
$$R_z = d_3 - (b_3)m_1 = A_3$$

Let A_1 be the largest,
$$A_1/d_1 = m_2$$
$$R_x = A_1 - d_1m_2 = B_1$$
$$R_y = A_2 - a_2m_2 = B_2$$
$$R_z = A_3 - a_3m_3 = B_3$$

Solution:
$$x = \Sigma \delta x$$
$$y = \Sigma \delta y$$
$$z = \Sigma \delta z$$

Numericals:

Q. Using SOR method, solve:

$$10x - 2y - 3z = 205$$
$$-2x + 10y - 2z = 154$$
$$-2x - y + 10z = 120$$

Soln: Here,

$$10x - 2y - 3z = 205$$
$$-2x + 10y - 2z = 154$$
$$-2x - y + 10z = 120$$

The residuals are obtained by

$$R_x = 205 - 10x + 2y + 3z$$
$$R_y = 154 + 2x - 10y + 2z$$
$$R_z = 120 + 2x + y - 10z$$

Let $x = y = z = 0$ be the initial guess.

Operation Table:

	R_x	R_y	R_z
$\delta x = 1$	-10	2	2
$\delta y = 1$	2	-10	1
$\delta z = 1$	3	2	-10

Relaxation Table:

	R_x	R_y	R_z
$x=y=z=0$	205	154	120
$\delta x=20.5$	0	195	161
$\delta y=19.5$	39	0	180.5
$\delta z=18.05$	93.15	36.1	0
$\delta x=9.32$	-0.1	54.7	18.6
$\delta y=5.5$	10.9	-0.3	24.1
$\delta z=2.4$	18.1	4.5	0.1
$\delta x=1.8$	0.1	8.1	3.7
$\delta y=0.8$	1.7	0.1	4.5
$\delta z=0.5$	3.2	1.1	-0.5
$\delta x=0.32$	0	1.74	0.14
$\delta y=0.174$	0.348	0	0.314

$\delta x=205/10=20.5$

$R_x=205-(10)\times20.5=0$

$R_y=154-(-2)\times20.5=195$

$R_z=120-(-2)\times20.5=161$

$\delta y=195/10=19.5$

$R_x=0-(-2)\times19.5=39$

$R_y=195-(10)\times19.5=0$

$R_z=161-(-1)\times19.5=180.5$

$\delta z=180.5/10=18.05$

$R_x=39-(-3)\times18.05=93.15$

$R_y=0-(-2)\times18.05=36.1$

$R_z=18.05-(10)\times18.05=0$

⋮ so on

Hence,

$x=\Sigma\delta x=20.5+9.32+1.8+0.32=31.94$

$y=\Sigma\delta y=19.5+5.5+0.8+0.174=25.974$

$z=18.05+2.4+0.5=20.95$

Hence by SOR method the solution of given system is;

$x=31.94\approx32$

$y=25.974\approx26$

$z=20.95\approx21$ Ans

2. Using SOR method solve;

$20x+y-2z=17$

$3x+20y-z=18$

$2x-3y+20z=25$

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Sol. Here,

$$20x + y - 2z = 17$$
$$3x + 20y - z = 18$$
$$2x - 3y + 20z = 25$$

The residuals are obtained by

$$R_x = 17 - 20x - y + 2z$$
$$R_y = 18 - 3x - 20y + z$$
$$R_z = 25 - 2x + 3y - 20z$$

Let $x = y = z = 0$ be the initial guess.

Operation Table:

	R_x	R_y	R_z
$\delta x = 1$	20	1	-2
$\delta y = 1$	3	20	-1
$\delta z = 1$	2	-3	20

Relaxation Table:

~~x=y=z=0~~	R_x	R_y	R_z
$x = y = z = 0$	17	18	25
$\delta z = 1.25$	-8	16.75	27.5
$\delta x =$			

$\delta z = 25/20 = 1.25$

$R_x = 17 - (-20) \times 1.25$
$= -8$

$R_y = 18 - (-1) \times 1.25$
$= 19.25$

$R_z = 25 - (+2) \times 1.25$
$= 27.5$

Operation Table:

	R_x	R_y	R_z
$\delta x = 1$	-20	-3	-2
$\delta y = 1$	-1	-20	3
$\delta z = 1$	2	1	-20

Relaxation Table:

	R_x	R_y	R_z
$x=y=z=0$	17	18	25
$\delta z = 1.25$	19.5	19.25	0
$\delta x = 0.975$	0	16.325	-1.95
$\delta y = 0.816$	-0.816	0.009	4.398
$\delta z = -0.220$	-0.376	0.225	0.002
$\delta x = -0.019$	0.004	0.282	0.04
$\delta y = 0.014$	-0.01	0.002	0.82
$\delta z = 0.041$	0.072	0.043	0

$\delta z = 25/20 = 1.25 = 1.25$

$R_x = 17 - (-2) \times 1.25 = 19.5$

$R_y = 18 - (-1) \times 1.25 = 19.25$

$R_z = 25 - 20 \times 1.25 = 0$

$\delta x = 19.5/20 = 0.975$

$R_x = 19.5 - (20) \times 0.975 = 0$

$R_y = 19.25 - (3) \times 0.975 = 16.325$

$R_z = 0 - (2) \times 0.975 = -1.95$

Hence,

$x = \Sigma \delta x = 0.975 - 0.019 = 0.956$

$y = \Sigma \delta y = 0.816 + 0.014 = 0.83$

$z = \Sigma \delta z = 1.25 - 0.220 + 0.041 = 1.071$ Ans

Power Method for the largest Eigen value and Corresponding Eigen vector:

⇒ Let A be the given square matrix. Let $X_0 = \begin{bmatrix}1\\0\\0\end{bmatrix}$ or $\begin{bmatrix}1\\1\\0\end{bmatrix}$ or $\begin{bmatrix}1\\1\\1\end{bmatrix}$ be the initial guess for the eigen vector.

Then,

$$A \cdot X_0 = X = \begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = x_2 \begin{bmatrix}x_1/x_2\\1\\x_3/x_2\end{bmatrix}$$; suppose x_2 be larges

$$= \lambda_1 \cdot X_1$$

$$A \cdot X_1 = X = \begin{bmatrix} \\ \\ \end{bmatrix} = \lambda_2 \cdot X_2$$

Numerical:

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1. Determine the largest eigen value and corresponding eigen vector using power method:

$$A = \begin{bmatrix}1 & -2 & 2\\4 & 4 & -1\\6 & 3 & 5\end{bmatrix}$$

Sol

Let $X_0 = \begin{bmatrix}1\\0\\0\end{bmatrix}$ be the initial guess for the eigen vector.

Now,

$$A \cdot X_0 = \begin{bmatrix}1 & -2 & 2\\4 & 4 & -1\\6 & 3 & 5\end{bmatrix} \cdot \begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}1\\4\\6\end{bmatrix} = 6\begin{bmatrix}0.1667\\0.6667\\1\end{bmatrix} = \lambda_1 \cdot X_1$$

$$A \cdot X_1 = \begin{bmatrix}1 & -2 & 2\\4 & 4 & -1\\6 & 3 & 5\end{bmatrix} \cdot \begin{bmatrix}0.1667\\0.6667\\1\end{bmatrix} = \begin{bmatrix}0.8333\\2.3336\\8.0003\end{bmatrix} = 8.0003\begin{bmatrix}0.1042\\0.2917\\1\end{bmatrix} = \lambda_2 \cdot X_2$$

$$A \cdot X_2 = \begin{bmatrix}1 & -2 & 2\\4 & 4 & -1\\6 & 3 & 5\end{bmatrix} \cdot \begin{bmatrix}0.1042\\0.2917\\1\end{bmatrix} = \begin{bmatrix}1.5208\\0.5836\\6.5003\end{bmatrix} = 6.5003\begin{bmatrix}0.2340\\0.0898\\1\end{bmatrix} = \lambda_3 \cdot X_3$$

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$$A \cdot X_3 = \begin{bmatrix} 1 & -2 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix} \cdot \begin{bmatrix} 0.2340 \\ 0.0898 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.0544 \\ 0.2952 \\ 6.6734 \end{bmatrix} = 6.6734 \begin{bmatrix} 0.3078 \\ 0.0442 \\ 1 \end{bmatrix} = \lambda_4 X_4$$

$$A \cdot X_4 = \begin{bmatrix} 1 & -2 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix} \cdot \begin{bmatrix} 0.3078 \\ 0.0442 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.2194 \\ 0.408 \\ 6.9794 \end{bmatrix} = 6.9794 \begin{bmatrix} 0.3180 \\ 0.0585 \\ 1 \end{bmatrix} = \lambda_5 \cdot X_5$$

$$A \cdot X_5 = \begin{bmatrix} 1 & -2 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix} \cdot \begin{bmatrix} 0.3180 \\ 0.0585 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.201 \\ 0.506 \\ 7.0835 \end{bmatrix} = 7.0835 \begin{bmatrix} 0.3107 \\ 0.0714 \\ 1 \end{bmatrix} = \lambda_6 \cdot X_6$$

$$A \cdot X_6 = \begin{bmatrix} 1 & -2 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix} \cdot \begin{bmatrix} 0.3107 \\ 0.0714 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.1679 \\ 0.5284 \\ 7.0784 \end{bmatrix} = 7.0784 \begin{bmatrix} 0.3063 \\ 0.0746 \\ 1 \end{bmatrix} = \lambda_7 \cdot X_7$$

$$A \cdot X_7 = \begin{bmatrix} 1 & -2 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix} \cdot \begin{bmatrix} 0.3063 \\ 0.0746 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.1571 \\ 0.5236 \\ 7.0616 \end{bmatrix} = 7.0616 \begin{bmatrix} 0.3054 \\ 0.0741 \\ 1 \end{bmatrix} = \lambda_8 \cdot X_8$$

Hence,

Largest eigen value = 7.0616 Ans

Eigen vector $= \begin{bmatrix} 0.3054 \\ 0.0741 \\ 1 \end{bmatrix}$ Ans

2. Determine the largest eigen value and corresponding eigen vector using power method.

i> $A = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix}$ ii> $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$

Sol Here,

i> $A = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix}$

Let $X_0 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ be the initial guess for the eigen vector.

Now,

$$A \cdot X_0 = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 26 \\ 4 \\ 2 \end{bmatrix} = 26 \begin{bmatrix} 1 \\ 0.1538 \\ 0.0769 \end{bmatrix} = \lambda_1 \cdot X_1$$

$$A \cdot X_1 = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0.1538 \\ 0.0769 \end{bmatrix} = \begin{bmatrix} 25.3076 \\ 1.4614 \\ 2.3076 \end{bmatrix} = 25.3076 \begin{bmatrix} 1 \\ 0.0577 \\ 0.0912 \end{bmatrix} = \lambda_2 X_2$$

$$A \cdot X_2 = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0.0577 \\ 0.0912 \end{bmatrix} = \begin{bmatrix} 25.2401 \\ 1.1731 \\ 2.3648 \end{bmatrix} = 25.2401 \begin{bmatrix} 1 \\ 0.0465 \\ 0.0937 \end{bmatrix} = \lambda_2 \cdot X_3$$

$$A \cdot X_3 = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0.0465 \\ 0.0937 \end{bmatrix} = \begin{bmatrix} 25.1259 \\ 1.1395 \\ 2.1588 \end{bmatrix} = 25.1259 \begin{bmatrix} 1 \\ 0.0454 \\ 0.0859 \end{bmatrix} = \lambda_4 \cdot X_4$$

$$A \cdot X_4 = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0.0454 \\ 0.0859 \end{bmatrix} = \begin{bmatrix} 25.2172 \\ 1.1362 \\ 2.3436 \end{bmatrix} = 25.2172 \begin{bmatrix} 1 \\ 0.0450 \\ 0.0929 \end{bmatrix} = \lambda_5 \cdot X_5$$

Hence, The largest eigen value is 25.2172 and corres-ponding the eigen vector is $\begin{bmatrix} 1 \\ 0.0450 \\ 0.0929 \end{bmatrix}$ Ans

ii) $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$

let $X_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ be the initial guess for the eigen vector.

Now,

$$A \cdot X_0 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \lambda_1 \cdot X_1$$

$$A\cdot X_1 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \lambda_2 \cdot X_2$$

$$A\cdot X_2 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \\ 3 \end{bmatrix} = 4 \begin{bmatrix} 0.75 \\ -1 \\ 0.75 \end{bmatrix} = \lambda_3 \cdot X_3$$

$$A\cdot X_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.75 \\ -1 \\ 0.75 \end{bmatrix} = \begin{bmatrix} 2.5 \\ -3.5 \\ 2.5 \end{bmatrix} = 3.5 \begin{bmatrix} 0.7143 \\ -1 \\ 0.7143 \end{bmatrix} = \lambda_4 \cdot X_4$$

$$A\cdot X_4 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 0.7143 \\ -1 \\ 0.7143 \end{bmatrix} = \begin{bmatrix} 2.4286 \\ -3.428 \\ 2.4286 \end{bmatrix} = 3.428 \begin{bmatrix} 0.7085 \\ -1 \\ 0.7085 \end{bmatrix} = \lambda_5 \cdot X_5$$

$$A\cdot X_5 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 0.7085 \\ -1 \\ 0.7085 \end{bmatrix} = \begin{bmatrix} 2.417 \\ -3.417 \\ 2.417 \end{bmatrix} = 3.417 \begin{bmatrix} 0.7073 \\ -1 \\ 0.7073 \end{bmatrix} = \lambda_6 X_6$$

Hence, the largest eigen value is 3.417 and the corresponding eigen vector is $\begin{bmatrix} 0.7073 \\ -1 \\ 0.7073 \end{bmatrix}$. Ans